

MNIST Generative Models Lab: From Manual Classifiers to VAE, DDPM, and GANs

Technical report (portfolio project)
mnist-generative-models-lab

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Abstract

I built a notebook-driven MNIST lab with two tracks: classification (from-scratch KNN and logistic regression through a PyTorch CNN) and generation (VAE, pixel-space DDPM, DCGAN, and a latent diffusion MLP). On the saved notebook outputs I kept, test accuracy climbs from **0.9705** (GPU KNN, $k=3$) to **0.9913** (comparison CNN), with logistic regression sitting lower at **0.9241** because it is strictly linear. The generative notebooks train end-to-end and produce readable digit samples, but I did not compute FID or Inception Score; my conclusions about sample quality are qualitative, based on the PNGs exported to **results/figures/**. What I learned most from this project is that a simple dataset still teaches real lessons about model capacity, inductive bias, and how generative families trade off blur, training cost, and sample sharpness. I also learned to document awkward results (the logistic dip, soft VAE reconstructions, duplicate training) instead of hiding them.

1 Introduction

I chose MNIST [2] because it is the standard place to make model behaviour visible before scaling up. This repo is not a production system; it is a curriculum I can walk through in Jupyter: understand the data, implement classical models by hand, add a small CNN, benchmark everything side by side, then explore compression (VAE), denoising (DDPM), adversarial training (DCGAN), and finally diffusion in a 16-D latent space.

The main thing I wanted from the lab was not a leaderboard score. I wanted to feel each step: write the distance function for KNN, implement the softmax gradient by hand, watch a CNN filter turn into an edge detector, then see how a VAE bottleneck softens images before a diffusion model tries to undo noise in a smaller space. Figure 1 shows the order I used. Each notebook preserves embedded outputs so metrics and plots are inspectable without retraining. All figures in this report were copied from those outputs into **results/figures/**; I did not regenerate checkpoints from gitignored **training_results/** or ***.pth** files.

MNIST lab learning path (classification → generative)

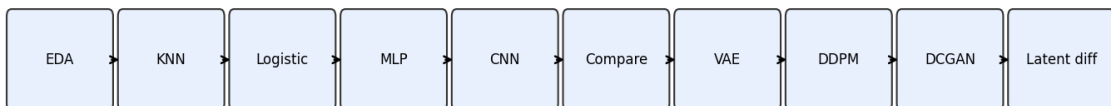


Figure 1: Learning path I followed: EDA, manual classifiers, CNN, multi-model comparison, then generative models.

2 Exploratory data analysis

Before training anything, `eda_analysis.ipynb` loads MNIST and walks through class balance, per-digit galleries, mean templates, pixel variance maps, intensity histograms, and PCA structure. On visual inspection, the class counts are nearly even (Figure 2a), which means raw accuracy is a reasonable metric and I do not need heavy rebalancing tricks for this dataset.

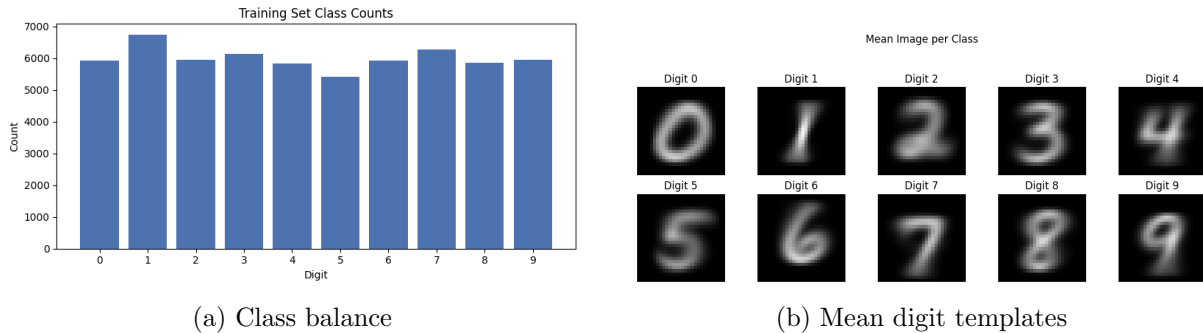


Figure 2: EDA findings from my saved notebook run. Digits are centered and stroke thickness varies by class.

The mean images (Figure 2b) look like the digits I expect, but PCA plots (Figure 3) show substantial overlap between classes in the first two components. That overlap stuck with me. MNIST is often called “easy,” yet a single linear boundary in pixel space still cannot separate all ten digits cleanly. I took that as early evidence that I would need non-linear models, which is exactly what the later notebooks tested.

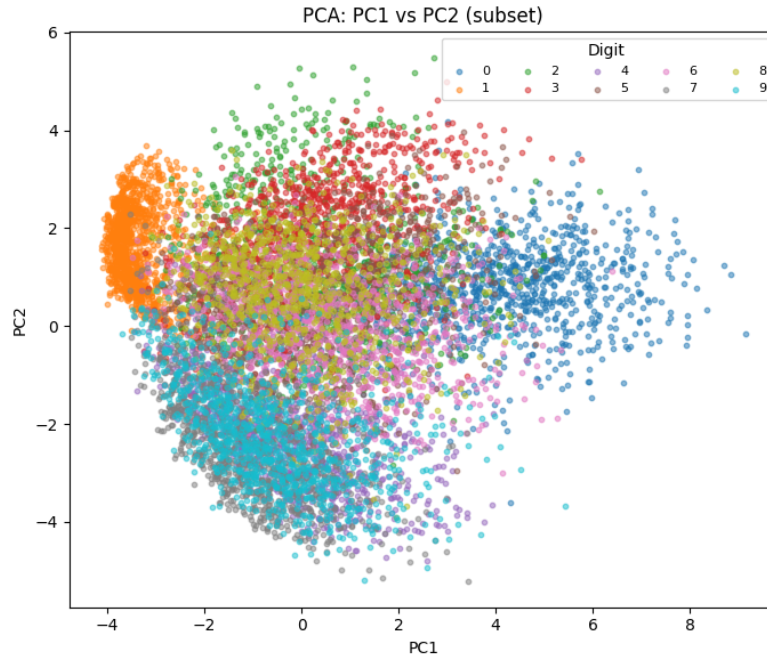


Figure 3: PCA structure in the training set. Classes overlap in low-dimensional projections, which foreshadows linear-model limits.

3 Classical and manual machine learning

3.1 KNN and the bias-variance tradeoff

In `manual_knn.ipynb` I implemented KNN in NumPy and then batched it on GPU with `torch.cdist`. A sweep over $k \in \{1, 3, 5, 9, 15, 20\}$ (Figure 4) shows accuracy peaking near $k=3$ at **0.9705** on the full test set. Very small k is noisier; very large k oversmooths decision boundaries. I found this plot more instructive than the final scalar accuracy because it makes the tradeoff concrete. I learned that even on MNIST, KNN is a strong baseline if you can afford the distance computation, and that tuning k is not cosmetic: it changes the kind of mistakes the model makes.

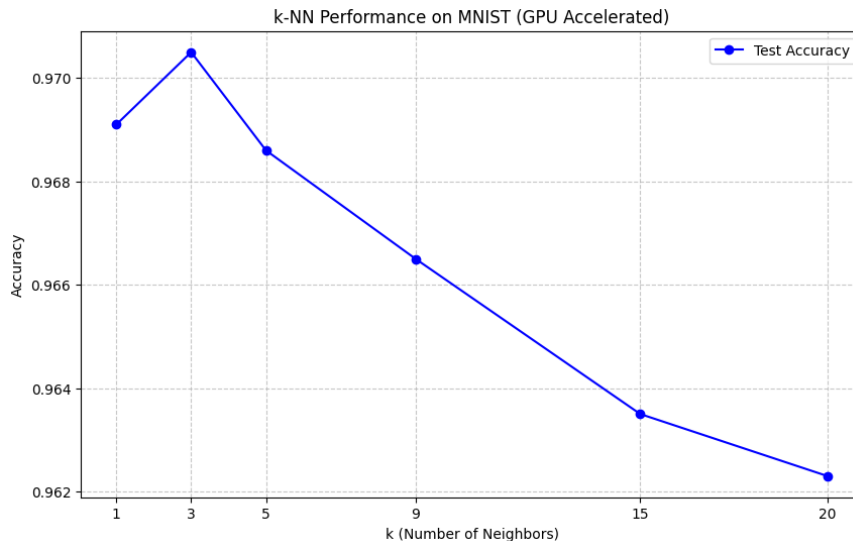


Figure 4: GPU KNN accuracy vs. k from my notebook outputs.

3.2 Logistic regression: a dip I did not expect

I expected accuracy to improve monotonically as models gained capacity. That did not happen. My from-scratch softmax regression in `manual_logistic_reg.ipynb` plateaus at **0.9241**, below KNN and well below the manual MLP. I think that is mostly a capacity issue: logistic regression is a linear classifier in pixel space, and MNIST digits written in different styles still overlap when flattened to 784 dimensions. Figure 5 shows the kind of errors I get, mostly confusions between visually similar strokes (for example 4 vs 9, 3 vs 5). The lesson for me was blunt: more “math in the notebook” does not automatically beat a simple non-parametric method if the model class is too restrictive.

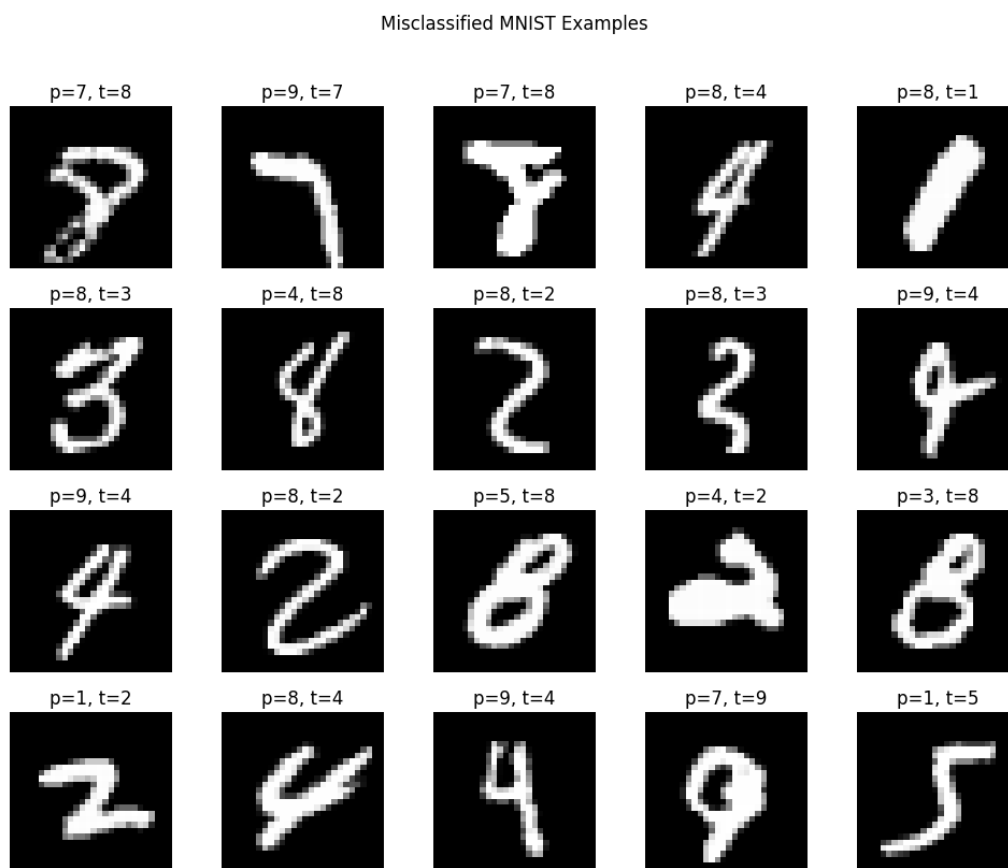


Figure 5: Misclassified test digits from my logistic regression run.

3.3 Manual MLP

Adding one ReLU hidden layer ($H=128$, 100 epochs) in `manual_mlp.ipynb` raised accuracy to **0.9794**. The first-layer weight templates (Figure 6) look like crude stroke detectors; that made backprop feel less abstract than reading equations alone. I learned that a single hidden layer is enough to recover most of the KNN gap, which helped me see MLPs as “learned feature extractors” rather than black boxes.

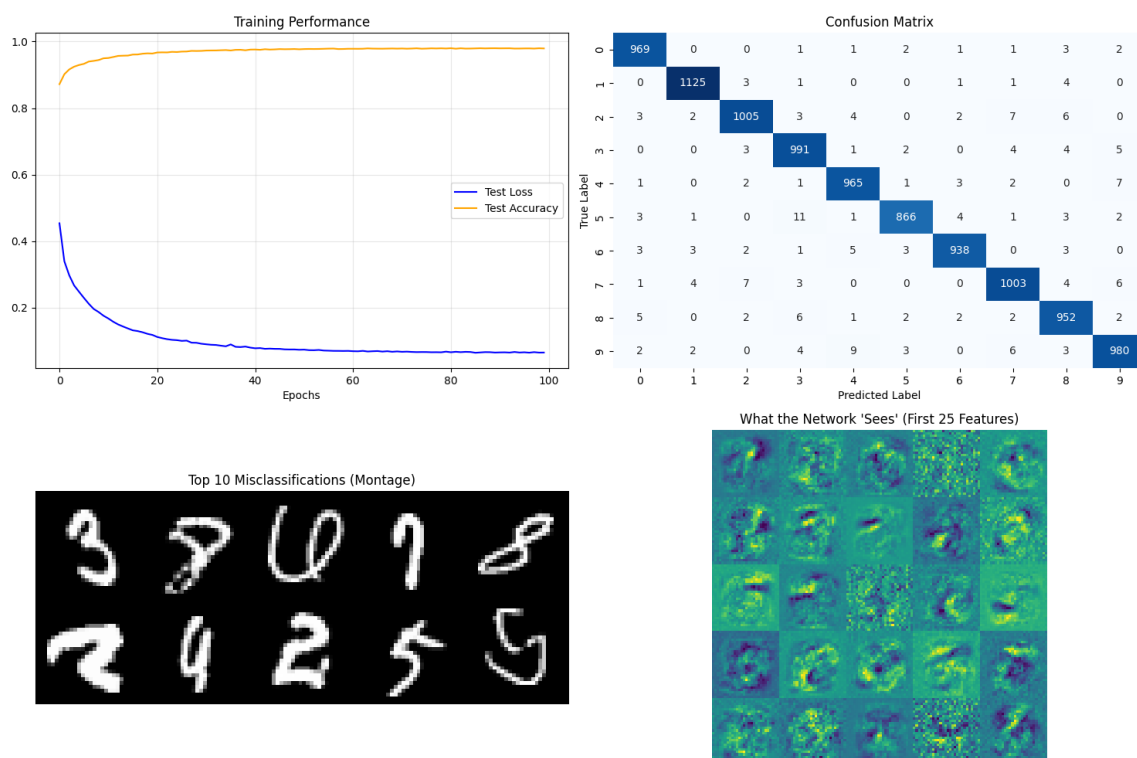


Figure 6: Manual MLP diagnostics from my run: loss/accuracy curves, confusion matrix, and first-layer filters.

Table 1 summarises the classification ladder through the manual notebooks and the standalone CNN.

Table 1: Classification metrics from saved notebook outputs (results/metrics/classification.json).

Model	Test accuracy	Source notebook
KNN (GPU, $k=3$)	0.9705	manual_knn.ipynb
Logistic regression	0.9241	manual_logistic_reg.ipynb
Manual MLP	0.9794	manual_mlp.ipynb
SimpleCNN	0.9871	cnn.ipynb
Comparison CNN	0.9913	02_mnist_models.ipynb

4 CNN baseline

`cnn.ipynb` implements a small `SimpleCNN`: two convolutional layers, max pooling, and two linear layers, trained five epochs with SGD. Test accuracy reaches **0.9871** by epoch 4. The jump from manual MLP to CNN is modest in absolute terms on MNIST, but I think the spatial inductive bias matters: convolutions reuse filters across the image instead of treating every pixel independently. Seeing the first-layer filters (Figure 8) line up with coursework examples made the inductive-bias argument feel real rather than theoretical.

Figure 7 shows my post-training analysis panel (confusion matrix, ROC-style curves, per-class behaviour). I learned to read confusion matrices as a story about remaining errors, not just a single accuracy number.

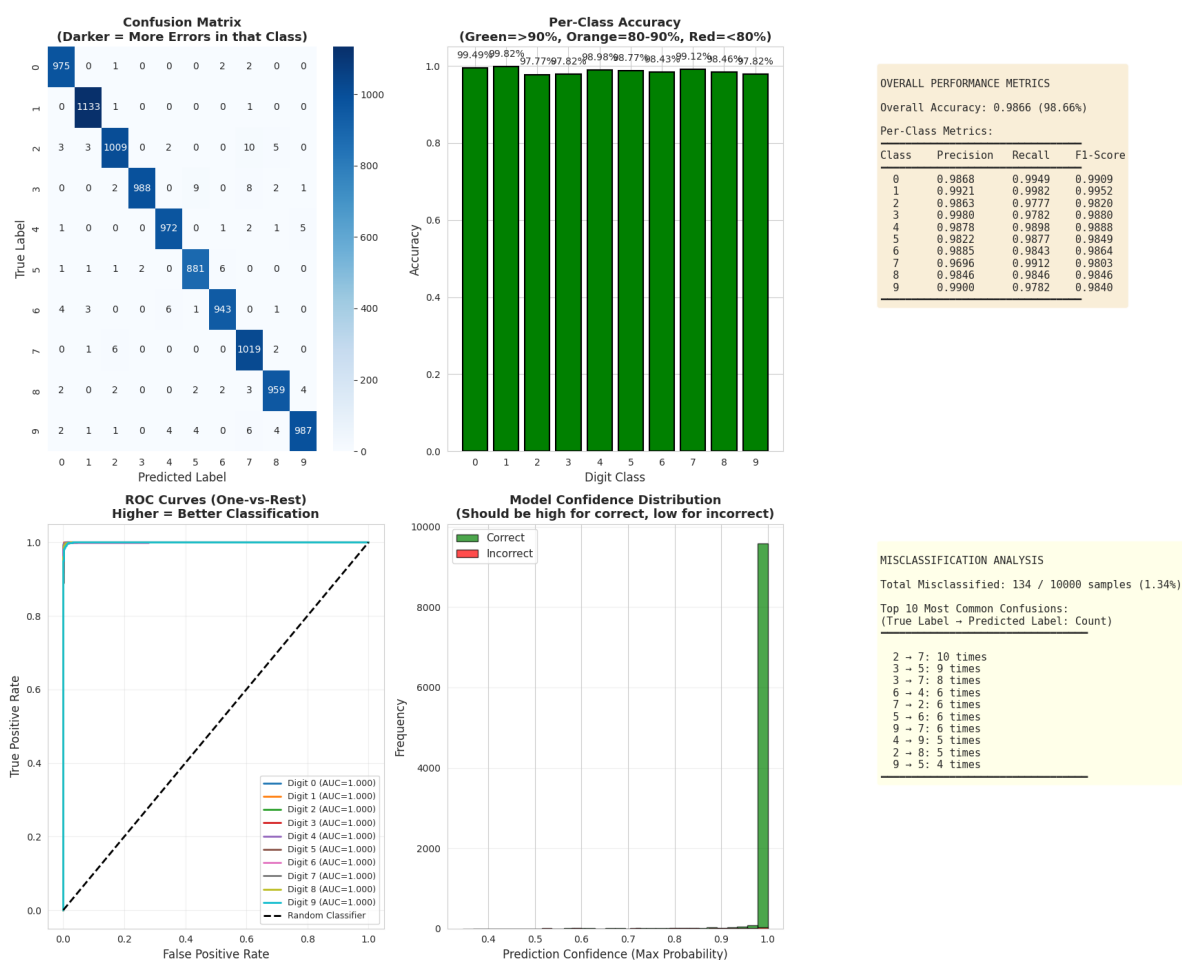


Figure 7: CNN evaluation panel from my saved run (confusion matrix and ROC-related plots in one figure).

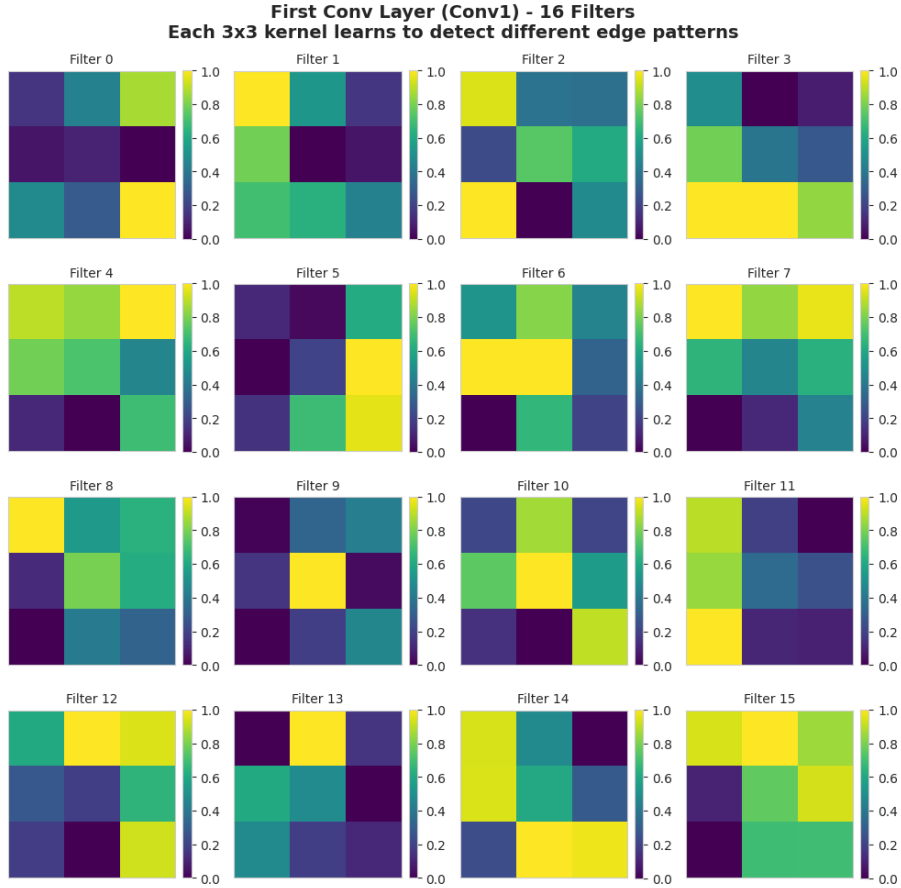


Figure 8: Learned filters in the first convolutional layer.

5 Generative models

After classification, I worked through four generative approaches in increasing complexity. I did not run FID or IS; I judged samples by eye on the PNG grids in `results/figures/`. That is a weakness for claiming generative quality, but it forced me to look carefully at what each family actually outputs.

5.1 Variational autoencoder

`vae.ipynb` trains a convolutional VAE with a 16-D latent space for 50 epochs. Final train/validation total loss is **100.22 / 100.09** (reconstruction plus weighted KL). Reconstructions (Figure 9a) preserve digit identity but look soft, which is what I usually see with Bernoulli/BCE or MSE-style decoders on MNIST. I learned that compression and sharpness pull in opposite directions: the VAE gives me a usable latent space, but I should not expect crisp samples from the decoder alone.

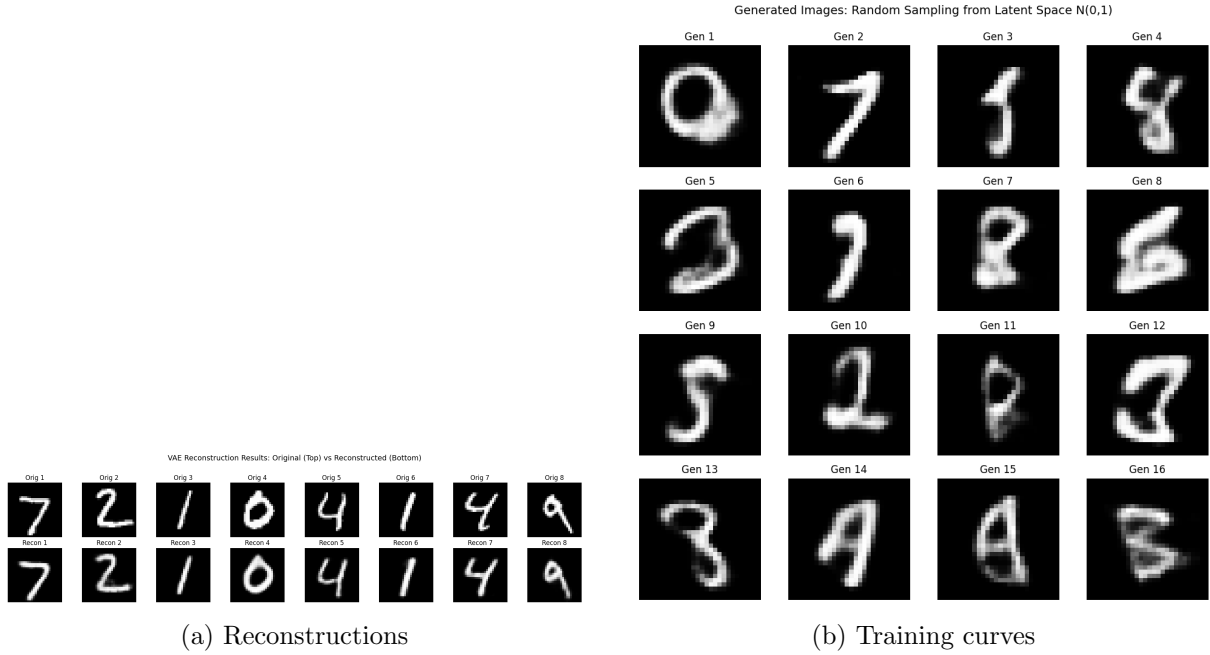


Figure 9: VAE outputs from `vae.ipynb`.

5.2 Pixel-space DDPM

`ddpm.ipynb` implements a DDPM [1] with $T=300$ timesteps and a U-Net denoiser, trained 15 epochs. Average loss at epoch 15 is **0.0454** (Figure 10a). Generated samples (Figure 10b) look like MNIST digits on inspection, though training in full 784-D pixel space felt slower and heavier than the latent diffusion stage later. I learned why latent diffusion exists: the math is elegant, but denoising every pixel step by step is expensive even on 28x28 images.

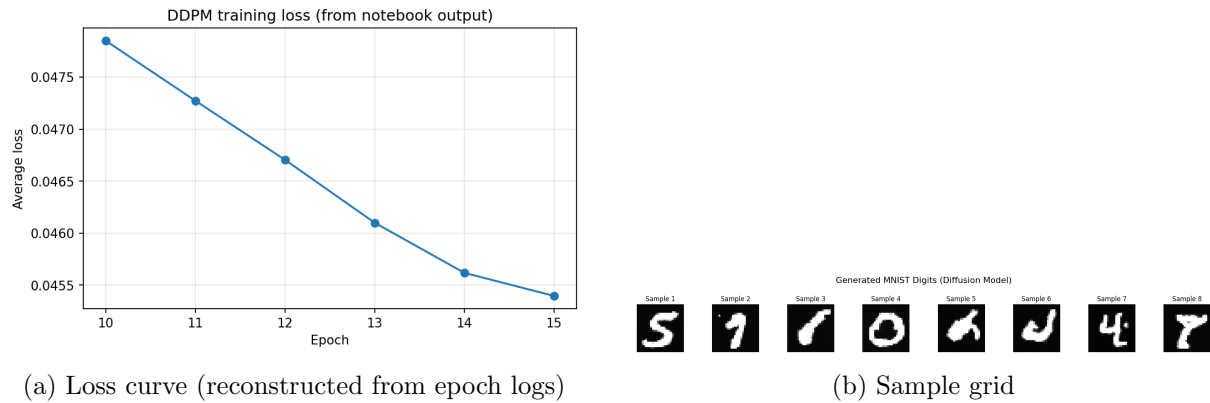


Figure 10: DDPM training and samples from my notebook outputs.

5.3 DCGAN

`drgan.ipynb` trains a DCGAN [3] for 50 epochs. Final generator/discriminator losses are **3.41** / **0.32**. The sample grid (Figure 11a) shows recognisable digits; interpolation (Figure 11b) suggests the latent space is at least locally smooth. I would not call GAN training stable here. The loss balance moved around, and I spent more time watching curves than I would like. Still, the saved outputs are usable for demonstration, and I learned that adversarial training can produce sharper digits than my VAE decoder, at the cost of trickier optimization.

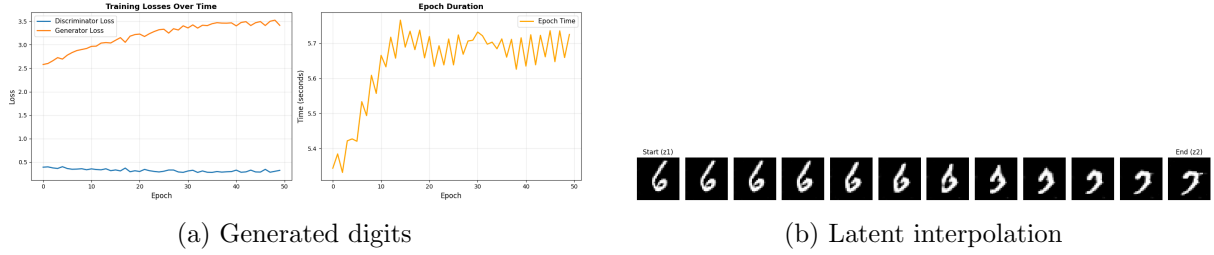


Figure 11: DCGAN outputs from my saved run.

5.4 Latent diffusion MLP

The notebook `latent_diffusion_mlp.ipynb` retrains a VAE, extracts latents, and trains a `LatentDiffusionMLP` for 150 epochs (MSE **0.224** near the end). Figure 12b shows reconstructions and generated samples from that notebook. Functionally, this is the pattern I later used on CIFAR at larger scale: compress first, diffuse second. I learned that reusing the VAE idea across notebooks is convenient while learning, but duplicating training is poor engineering hygiene.

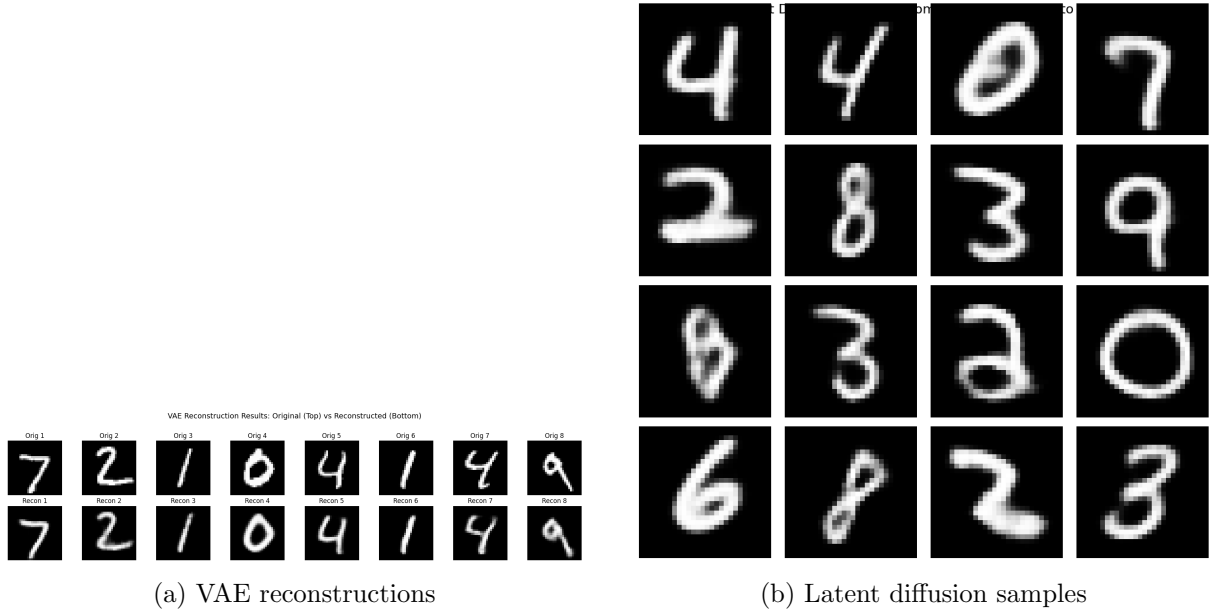


Figure 12: Latent diffusion track. Samples are digit-like on visual inspection; sharpness is limited by the VAE bottleneck.

6 Comparison and insights

`02_mnist_models.ipynb` benchmarks dummy, sklearn, and PyTorch models on a shared harness. The sorted accuracy bar chart (Figure 13a) makes the ladder obvious: deep learning wins on MNIST, but classical KNN is already strong, and the gap from MLP to CNN is incremental rather than dramatic. The comparison CNN reaches **0.9913**; remaining errors cluster on visually similar pairs (Figure 13b).

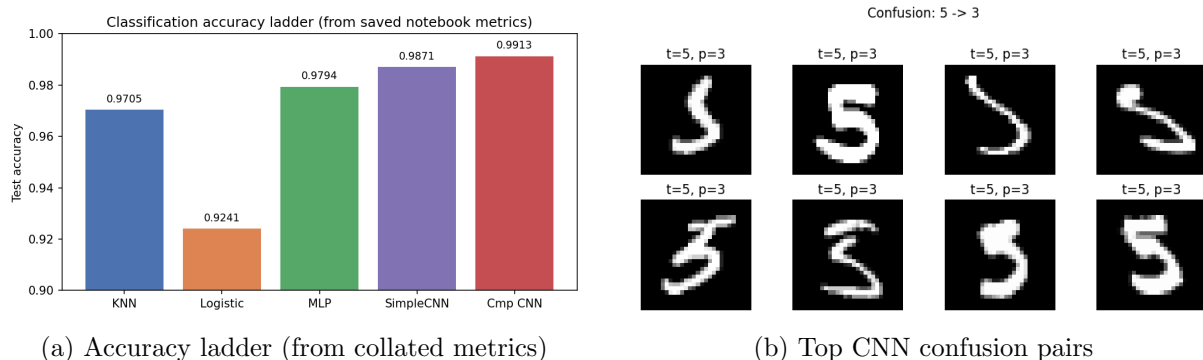


Figure 13: Multi-model comparison insights from my saved outputs.

On the generative side, I now think of the tradeoffs this way: VAEs give cheap latent codes but blurry decoders; pixel DDPMs are faithful to the DDPM math but expensive in 784-D; GANs can look sharp but need careful monitoring; latent diffusion is efficient once latents exist but inherits VAE quality. I did not run controlled ablations to rank them quantitatively, so this is personal takeaway from the PNGs, not a benchmark result.

7 What I learned

Looking back at the full lab, a few lessons stand out:

- **Start with the data.** EDA took time, but it explained later failures (linear models struggling, confusion pairs that look like handwriting variation).
- **Implement before abstracting.** Manual KNN, logistic regression, and MLP code made later PyTorch models easier to reason about.
- **Accuracy ladders lie a little.** Logistic regression at 0.9241 is not a bug; it is a reminder that model class matters as much as tuning.
- **Generative models teach different skills.** VAEs taught compression and ELBO trade-offs; DDPMs taught noise schedules; GANs taught unstable co-training; latent diffusion tied the threads together.
- **Document the rough edges.** Duplicate VAE training, unpinned dependencies, and missing FID scores are part of the honest story of how I built this repo.

8 Limitations

- **No FID / IS.** I did not compute standard generative metrics. Claims about sample quality are qualitative.
- **28x28 grayscale only.** Findings may not transfer to color or high-resolution data.
- **Duplicate VAE training.** `vae.ipynb` and `latent_diffusion_mlp.ipynb` both train a VAE; I should load a checkpoint instead of retraining.
- **Notebook hygiene.** `02_mnist_models.ipynb` may list duplicate CNN rows from re-runs; I cite **0.9913** as the comparison result.

9 Future work

- Deduplicate the VAE pipeline (train once, load `vae_model.pth` elsewhere).
- Add FID or a simple classifier-based generative score.
- Guard Colab-only cells for local runs; dedupe VAE training across notebooks.
- Extend the same notebook structure to Fashion-MNIST.

- Add a lightweight `nbconvert` smoke test in CI.

10 Conclusion

I built an end-to-end MNIST lab that reads like a course project I actually ran: EDA, manual implementations, a CNN benchmark, and four generative families with saved plots in `results/`. Classification reaches **0.9913** on the comparison CNN; generative models produce inspectable digit samples without me claiming state-of-the-art numbers. The part I am most glad I kept is the reflective thread: the logistic-regression dip, the soft VAE reconstructions, the heavy pixel DDPM, and the missing FID metrics. Those details explain what I learned, not just what I achieved.

References

- [1] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In *Advances in Neural Information Processing Systems*, volume 33, pages 6840–6851, 2020.
- [2] Yann LeCun, Léon Bottou, Yoshua Bengio, and Patrick Haffner. Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11):2278–2324, 1998.
- [3] Alec Radford, Luke Metz, and Soumith Chintala. Unsupervised representation learning with deep convolutional generative adversarial networks. *arXiv preprint arXiv:1511.06434*, 2016.